“Dump my brain so it remembers for me”: An AI-Augmented External Autobiographical Memory for Aphantasia and SDAM

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This is the accessible version of the printed poster. It contains all text on the poster, a description of the diagram, and the longer sentences that were shortened to fit the print.

- Printed poster (A0 PDF, visual layout)
- Paper: doi.org/10.1145/3797867.3841206 (resolves once the ACM proceedings are published)

Key message

> **I supply intent. The agent owns the rest of the memory loop: it gathers, connects, indexes, and surfaces, because I will not pull.**

My profile

Aphantasia: no voluntary mental imagery. **SDAM** (severely deficient autobiographical memory): I cannot relive my own past. Both are self-identified, not diagnosed.

Survey of Autobiographical Memory scores, out of 5

Domain	Score
Episodic	1.25
Future	1.17
Spatial	4.17

VVIQ-16: 21 of 80 (aphantasia range). One faculty is high, the rest are at the floor.

Why memory aids fail me

- **Nothing tells me what to record.** Faced with a blank diary, I stall.

- **I never return to read it.** The record, and the memory with it, withers.
- **Visual aids form no internal image**, and I rarely take photos.

The fix is not more storage but a system that elicits what to capture and returns it when it matters.

What my profile changes in the design

The poster lists the first three rows. Each row is a design choice that would be a mistake for a typical user but is required for this profile.

Design choices that invert for this profile

Trait	So the design	For a typical user
No imagery (aphantasia)	Works in words and numbers, not pictures; images only illustrate.	Photos and visual capture cue memory for later review.
No reconnecting in my head (SDAM)	The agent does the connecting across sources and time.	Fragmented context is a tax paid in toggling; the reconnecting still happens in the head.
No baseline to lose	It pushes, even unasked; an un-pushed item is simply lost.	Push costs attention, and offloading risks disengagement.
I never return to read	The path where I file or browse the store myself is removed.	A person files each item and builds their own index.
No internal check	I recap aloud so the agent can confirm or correct; how I remember feeling is not stored.	A person can weigh a claim against their own hazy memory.

How the system works (diagram)

The poster shows a flow diagram with three boxes from left to right: Me (supply intent); AI agent, the largest box (sole reader, writer, organizer; gather, connect, index, surface); and Governed store (rules plus index). A Sources box at the top (messages, work, AI chats, calendar, health) feeds the store through an arrow labeled auto-ingest. Me sends intent to the agent and discusses with it; the agent pushes and surfaces context back to me. The agent records to and indexes the store, and reads the store first before acting. A dashed path from Me directly to the store, labeled “file or browse the store myself”, is marked as removed.

A loop I do not operate. The agent is the store's only reader, writer, and organizer. The general-purpose agent (Claude) runs interactively and on a schedule, guided by per-folder rule files.

One month of use (2026-05-20 to 2026-06-20)

- **29 of 32** days with a captured daily entry
- **46** meeting notes (16 auto-pulled)
- **174** logged AI working sessions
- **27** items filed unconfirmed, **5** held back as too ambiguous to place

Descriptive counts from one participant, not efficacy measures. One auto-pulled meeting was misfiled, caught only at weekly review.

Four design principles from daily use

P1, the linchpin: Proactive recall and cross-source connection

Before a meeting, the agent assembled, unasked, the last outcome, open action items, and the messages in between: context from four or five places.

P2: Language and number as the channel

Long, disfluent, mixed-language speech goes in; prose and numbers come out. I read progress off data, such as knee flexion over weeks.

P3: Closure by articulation

I recap aloud; the agent confirms or corrects. One recap caught three misreadings in action items that had looked complete.

P4: Resurfacing dispersed threads

A question raised, set aside, and reopened over weeks was rebuilt into what was settled, rejected, and still open.

Connection is where the deficit bites

For a typical worker, fragmented context is a tax paid in toggling; the reconnecting still happens in the head. Mine does not, so the prosthesis must supply it. One risk does not invert: a confidently wrong output I cannot check against my own memory.

Limits

- N = 1, designer as participant.
- Needs a high digital footprint; a four-day trip cut capture to a fifth.
- Consent wins over coverage: nothing is recorded at home.

At the poster

I record my conversations here, audio only, to help me remember each day. If you mind, just tell me.

Poster C08 at ASSETS 2026, the 28th International ACM SIGACCESS Conference on Computers and Accessibility. All content is drawn from the paper.